

Jin Tan Ruan

Software Developer · Forward Deployed Engineer

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AI Forward Deployed Engineer with 5+ years building and deploying production generative AI systems in enterprise environments, including multimodal video AI platforms processing 10M+ assets. Specialized in multimodal AI, multi-vector embeddings, RAG architectures, agent-based systems, and customer-embedded system design at scale across AWS and GCP.

Skills & Certifications

AI Systems & Architecture	Multimodal AI systems, RAG architectures, agent-based architectures, embedding pipelines, vector search, model fine-tuning (LoRA/QLoRA), inference optimization, evaluation & benchmarking
Programming & Systems	Python, Go, TypeScript, Java, C++, SQL, Bash; distributed systems, concurrency, API design
Cloud & Infrastructure	AWS, Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, CDK, CI/CD pipelines, production monitoring and observability
Data & Storage	PostgreSQL, MongoDB, Redis, Elasticsearch; vector databases (Pinecone, Weaviate, Qdrant); data modeling and indexing
Certifications	AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect Professional
Languages	English, Spanish (Native), Chinese (Native)

Work Experience

TwelveLabs

Senior Forward Deployed Engineer (AI Systems)

New York City, NY

Aug. 2025 – Present

- Served as technical lead for enterprise video AI deployments, designing and embedding production-grade multimodal systems within customer environments to support large-scale search, classification, and analytics workloads.
- Owned the architecture and operation of end-to-end video AI pipelines—including ingestion, embedding generation, indexing, and retrieval—powering mission-critical production workflows processing 10M+ video assets.
- Set technical direction for customer-facing integrations and reference architectures by designing scalable SDKs, APIs, and processing services, leading tradeoffs across model accuracy, throughput, and infrastructure cost, and standardizing deployment patterns that influenced platform capabilities and product roadmap priorities.
- Owned production video AI systems deployed across multiple global enterprises (Netflix, Disney, NFL, Roku, Protégé, Lionsgate), acting as a trusted technical advisor and meeting strict SLAs for availability, performance, and cost efficiency.

Amazon Web Services (AWS)

Senior Generative AI Prototyping Engineer

New York City, NY

Jan. 2022 – Aug. 2025

- Led the design and delivery of enterprise-grade generative AI reference architectures across healthcare, automotive, and media industries, leveraging RAG, LLMs, and agent-based systems to address complex production requirements.
- Played a foundational role in the launch of Amazon Bedrock agent capabilities by defining API schemas, implementing core orchestration workflows, and establishing best practices for customer adoption.
- Designed and open-sourced six generative AI solutions—including RAG pipelines, multimodal and agentic chatbots, and image generation systems—driving developer adoption and influencing ecosystem standards.
- Authored four technical whitepapers and blogs on prompt engineering and LoRA-based fine-tuning, shaping customer architecture decisions and informing internal product strategy.
- Partnered across product, research, and field organizations to translate early customer signals into scalable platform capabilities, directly influencing service roadmap and feature prioritization.

Education

Syracuse University — School of Engineering and Computer Science

M.S. & B.S. in Computer Science and Engineering, GPA: 3.7 (Cum Laude)

Syracuse, NY

Aug. 2016 – May 2021

- Selected for Concord Dawn — U.S. Air Force Cybersecurity Program (competitive national selection).

Academic Research

ICML (International Conference on Machine Learning)

Mood Swings Agents: Dynamic Emotional Adaptation in Multi-Agent Systems

New York City, NY

2025

- Proposed a novel multi-agent framework enabling dynamic emotional state adaptation based on environmental stimuli and inter-agent interactions.
- Designed and evaluated adaptive emotional transition algorithms balancing exploration–exploitation tradeoffs, demonstrating a 34% improvement in multi-agent coordination performance over static-agent baselines in non-stationary environments.
- Published peer-reviewed findings at ICML, contributing to research on agent autonomy, coordination, and adaptability in dynamic systems.